

REFINERY TECHNICAL SERVICES · EQUIPMENT INSPECTION REPORT

Report Ref: MRPL-INSP-2026-8842 | Date: 2026-08-20

Unit: Primary Refining Unit 02 - Flare Header & Pump Skid P-102A

Inspected By: Senior Maintenance Inspector

1. EXECUTIVE SUMMARY

A comprehensive visual and ultrasonic thickness inspection was carried out on Primary Refining Unit 02. Critical anomalies were detected requiring immediate engineering compliance evaluation.

2. DETAILED OBSERVATIONS & FINDINGS

- Finding 1: Severe galvanic corrosion fatigue detected on primary discharge flange bolts (Flange FL-102B). Ultrasonic thickness measurement indicates 1.65mm localized wall thinning, exceeding the 1.50mm critical limit.
- Finding 2: Pressure Relief Valve PRV-204 tag indicates last calibration was performed 14 months ago. Standard operating limits mandate maximum 12-month recalibration intervals for hydrocarbon service.
- Finding 3: Pump P-102A mechanical seal barrier fluid reservoir shows visible weeping of hydrocarbon fluid. Secondary containment pressure has dropped from 2.5 bar to 1.1 bar over a 48-hour operating window.

3. INSPECTOR CONCLUSION

Immediate formal approval note required to determine shutdown or remediation scheduling.

EQUIPMENT INSPECTION REPORT · APPENDIX A: MEASUREMENT DATA

Table 1: Flange FL-102B Ultrasonic Thickness Grid

Location	Nominal (mm)	Measured (mm)	Loss (mm)	Status
Bolt 1	14.00	12.35	1.65	CRITICAL DEFECT
Bolt 2	14.00	12.40	1.60	CRITICAL DEFECT
Bolt 3	14.00	13.10	0.90	MONITOR
Bolt 4	14.00	13.05	0.95	MONITOR

Table 2: PRV-204 Calibration Log

Tag ID	Service Type	Last Recert	Expiration Date	Status
PRV-204	Hydrocarbon	2025-06-10	2026-06-10	OVERDUE (2 Months)

End of Inspection Report MRPL-INSP-2026-8842